

FYS4480/9480 September 18

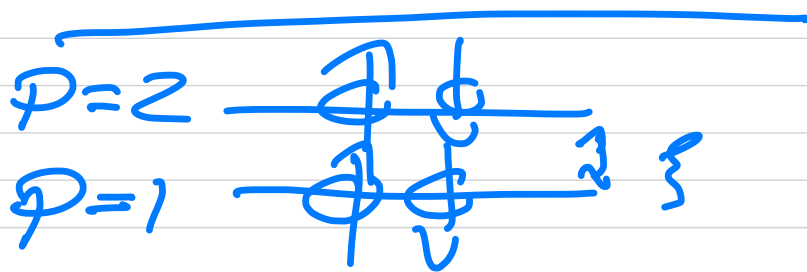


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$$n = 8 \quad 1 \quad N = 4$$

$$P=4 \text{ ---}$$

$$P=3 \text{ ---}$$



$$\} | \Phi_0 \rangle = a_{1+}^{\dagger} a_{1-}^{\dagger} a_{2+}^{\dagger} a_{2-}^{\dagger} \times | 0 \rangle$$

$$H = \sum_{p \neq} (p-1) a_{p+}^{\dagger} a_{p-}$$

$$- \frac{g}{2} \sum_{p \neq} a_{p+}^{\dagger} a_{p-}^{\dagger} a_{q-} a_{q+}$$

$$\langle p+p- | v | q+q- \rangle = -g/2 i \frac{2\pi\omega}{2\pi\omega}$$

$$\langle \Phi_0 | H | \Phi_0 \rangle = E_{ref}$$

$$H_0 = \sum_{p \neq r} (p-1) \langle 0 | a_{2-} a_{2+} a_{1-} a_{1+} \overbrace{a_{p-}^+ a_{p+}^+}^{a_{1+}^+ a_{1-}^+ a_{2+}^+ a_{2-}^+} \rangle$$

$\underbrace{\hspace{10em}}_{\times |0\rangle}$

$$= 2\xi \quad (\xi = 1)^{\delta_{p+}}$$

$(\hat{1}\hat{1})$ H_I $-\frac{g}{2} \sum_{p,q}$ $\langle 0 | a_{2-} a_{2+} a_{1-} a_{1+} a_{p+}^{\dagger} a_{p-}^{\dagger} a_{q-}^{\dagger} a_{q+}^{\dagger}$ $\times a_{1+}^{\dagger} a_{1-}^{\dagger} a_{2+}^{\dagger} a_{2-}^{\dagger} | 0 \rangle$ $= -g$ $E_{\text{ref}} = 2 - g$

$$\langle \Phi_0 | H | \Phi_i \rangle; \quad \langle \Phi_j | H | \Phi_i \rangle$$

$$\langle \Phi_0 | H | \Phi_1 \rangle$$

$$P=3 \quad \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$$

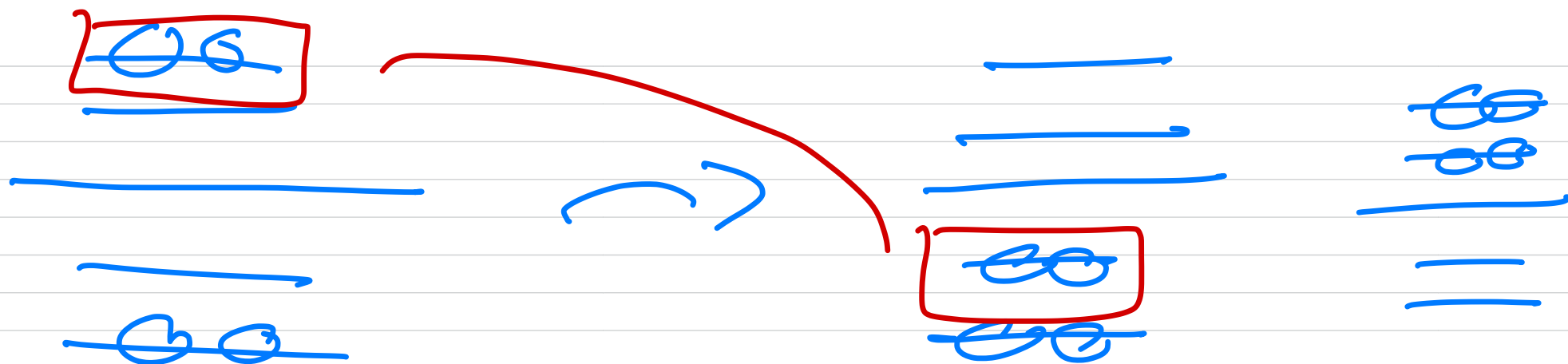
$$|\Phi_1\rangle \rightarrow$$

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$$

$$P=1 \quad \text{---}$$

$$\begin{array}{c} \boxed{a_{2-} a_{2+}} \quad \boxed{a_{1-} a_{1+}} \quad \boxed{a_{p+}^+ a_{p-}^+} \quad \boxed{a_{g-} a_{g+}} \quad \boxed{a_{1+}^+ a_{1-}^+} \quad \boxed{a_{3+}^+} \\ \times a_{3-}^+ \end{array}$$

$= - \quad g/2$



$1\Phi_1 >$

$1\Phi_2 >$

$0p04$

$2p24$

$2p24$

$2p24$

$2p24$

$4p44$

$0p04$

$2-9$

$-5/2$

$-5/2$

$-9/2$

$-9/2$

0

$2p24$

$-5/2$

$4-9$

$6-9$

0

0

$2p24$

$-5/2$

-9

0

$6-9$

$2p24$

$-5/2$

0

$8-9$

$2p24$

$-5/2$

0

$10-9$

$4p44$

0

Comparison with MBPT
to 2nd

$$\Delta E^{(2)} = \sum_{\substack{\Phi_1 \neq \\ \Phi_0}} \frac{|\langle \Phi_0 | H_1 | \Phi_1 \rangle|^2}{E_0^{(0)} - E_1^{(0)}}$$

$$= -\frac{7}{24} g^2$$

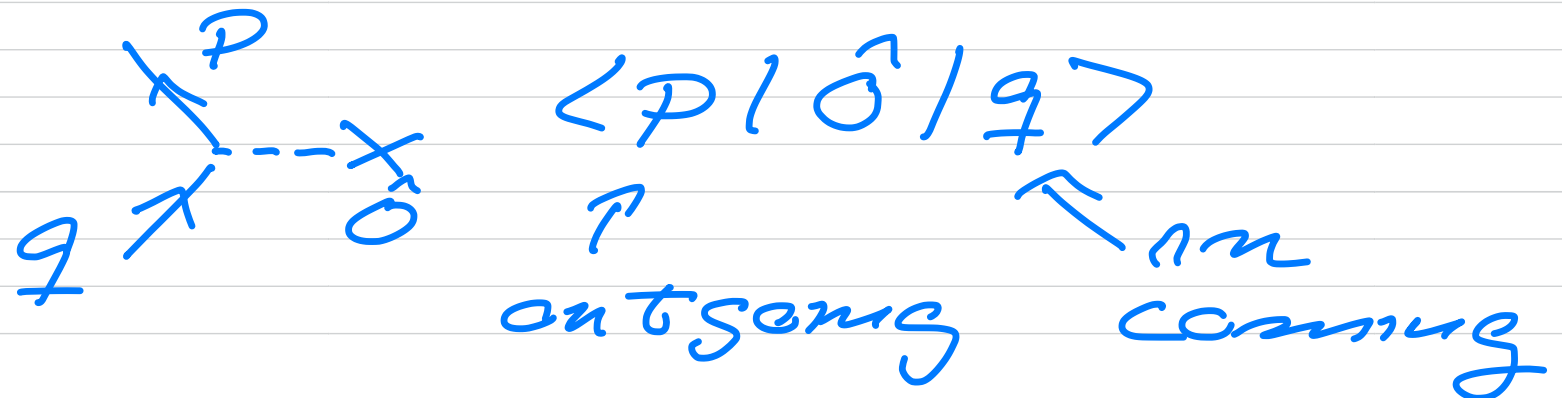
$$E_{\text{ref}} - \frac{7}{24} g^2 = 2 - g - \frac{7}{24} g^2$$

Diagrams

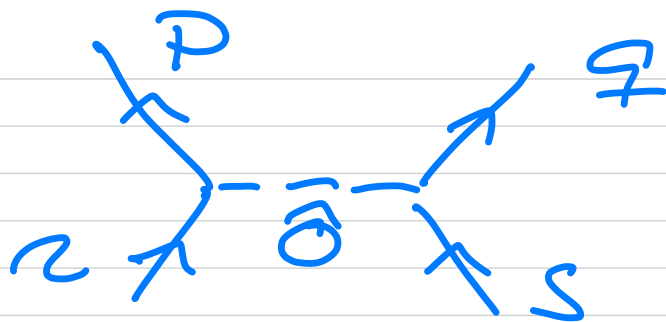
(i) particle states as upgoing arrows $\uparrow a$

(ii) hole states as downgoing arrows $\downarrow i$

(iii) one-particle operator



(i.i)



$$\langle p q | \hat{O} | r s \rangle$$

$$\langle \overbrace{\text{left cut}}^p \overbrace{\text{right cut}}^q | \hat{O} |$$

$$| \underbrace{\text{left cut}}_r \underbrace{\text{right cut}}_s \rangle$$

$$\Delta E = \sum_{ai} f_{ai} C_i^a + \frac{1}{4} \sum_{i'j'} \langle ab | h | i'j' \rangle C_{i'j'}^{ab}$$

$$f_{ai} \rightarrow \begin{array}{c} a \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} x \\ \diagup \quad \diagdown \\ i \end{array} = \begin{array}{c} a \quad i' \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} x \end{array}$$

$$\langle a | f | i \rangle$$

$$f_{ia} \rightarrow \begin{array}{c} \text{---} \text{---} \text{---} x \\ \diagup \quad \diagdown \\ a \quad i' \end{array}$$

$$\sum_{ai} f_{ia} c_i^a \rightarrow \begin{array}{c} \text{---} \text{---} \text{---} x \\ \diagup \quad \diagdown \\ a \quad i' \end{array}$$

$$c_i^a = \begin{array}{c} a \quad i' \\ \diagdown \quad \diagup \\ \text{---} \end{array} = \boxed{\begin{array}{c} \text{---} \text{---} \text{---} x \\ \diagup \quad \diagdown \\ \text{---} \end{array}}_{|p|h}$$

$$\sum_{\substack{p, q, \\ a, \lambda}} \langle \Phi_0 | a_p^\dagger q_q a_a^\dagger q_{\lambda'} | \Phi_0 \rangle \times \langle p | q | \lambda \rangle C_\lambda^a$$

a p14 state

$$= \sum_{q_i} f_{ia} C_\lambda^a$$

$$= \text{coop}$$

(v) $m_h = 1$
 $m_e = 1$

$$(-1)^{m_h + m_e} = (-1)^2 = +1$$

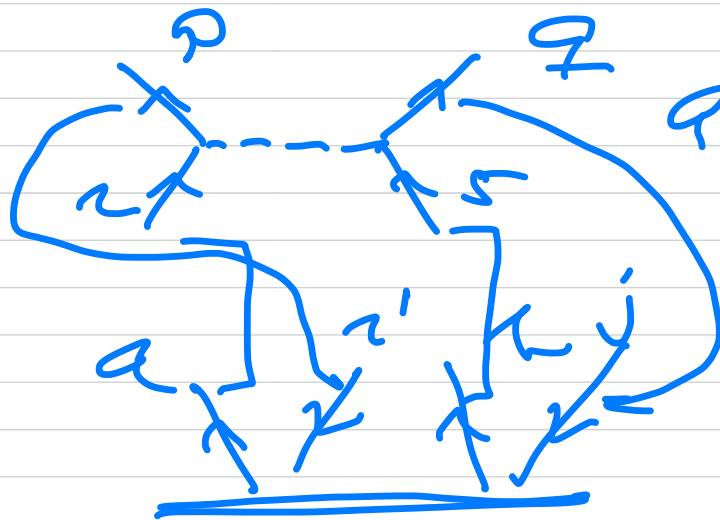
(VI) sum over all lines between vertices

$$\sum_{q_i} c_i^a |\Phi_i^a\rangle = \sum_{q_i'} c_i^a \underbrace{a_a^\dagger a_{i'}}_{\text{line}} |\Phi_i^a\rangle$$

$$\frac{1}{4} \sum_{\substack{ab \\ i'j'}} \langle ij | \psi | ab \rangle c_{i'j'}^{ab}$$

$$\frac{1}{4} \sum_{\substack{pqrs \\ ab \\ i'j'}} \langle \Phi_c | a_p^\dagger a_q^\dagger a_r a_s a_a^\dagger a_b^\dagger a_j a_i | \Phi_c \rangle \times \langle pq | \psi | rs \rangle_{AS} c_{i'j'}^{ab}$$

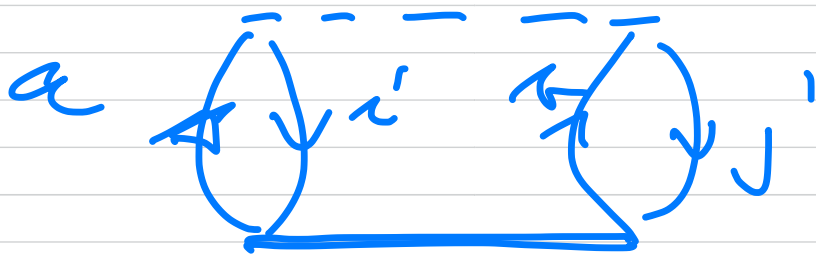
$$= \sum_{ab, i'j'} \langle i'j | \psi | ab \rangle C_{i'j}^{ab}$$



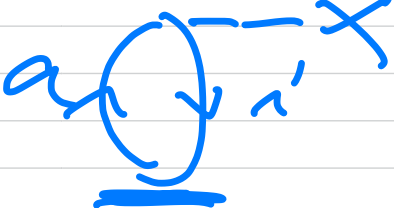
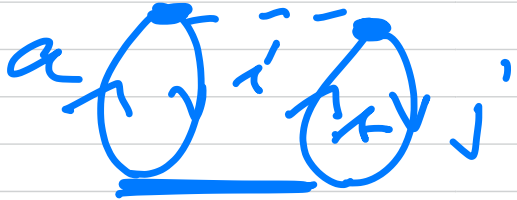
$$q_p^+ q_q^+ q_s q_r$$

$$\times \langle p q | \psi | r s \rangle_{A_5}$$

$$q_a^+ q_b^+ q_j q_{i'} \\ \times C_{i'j}^{ab}$$



$$\Delta E = \underbrace{a}_{1p1k} \underbrace{\bar{i} \bar{i}'}_{\text{diagram}} + a \underbrace{\bar{i} \bar{i}'}_{2p2k} \underbrace{\text{diagram}}_{\text{diagram}}$$

From FCI to approximation

1 Hartree-Fock theory

$$\hat{H}^{HF} |p\rangle = \epsilon^{HF} |p\rangle \quad \text{one body}$$

$$\hat{H}^{HF} = \hat{h}_0 + \hat{U}^{HF}$$

Φ_0^{HF} = Slater det
original basis Φ_0

$$E_{ref} = \langle \Phi_0 | H | \Phi_0 \rangle$$

$$E_{ref}^{HF} = \langle \Phi_0^{HF} | H | \Phi_0^{HF} \rangle$$

$$E_{ref}^{HF} \leq E_{ref}$$

$$\Phi_0^{HF} = \det[C] \Phi_0$$

$\nearrow$
mtx for unitary
transf

$$|\phi\rangle = \sum_{\lambda} c_{\phi\lambda} |\lambda\rangle$$

$\uparrow$ new basis $\uparrow$ original sp-basis

$$\langle\phi|U^{HF}|\phi\rangle = \sum_{i \leq F} \langle\phi i|v|\phi i\rangle_{AS}$$

$$\hat{h}_0 = \hat{t} + \hat{u}_{ext}$$

$\uparrow$
 kinetic energy

